

Prosimians

PHYLUM	Chordata
CLASS	Mammalia
ORDER	Primates
SUBORDER	Strepsirrhini
FAMILIES	4
SPECIES	139

More primitive than monkeys and apes, prosimians comprise the lemurs of Madagascar, the galagos and pottos of Africa, and the lorises of Asia. Lemurs (which include sifakas, the indri, and the aye-aye) have large ears, an elongated body, long limbs, and most have a long, bushy tail. Lorises, pottos, and galagos are generally smaller than lemurs and tend to have larger eyes. Prosimians mostly inhabit forests and are usually nocturnal (some lemurs are diurnal). Deforestation has endangered many prosimians, especially the lemurs.

Anatomy

Prosimians have a sense of smell that is more highly developed than in other primates. They have large eye sockets and a crystalline layer behind the retina of the eye that reflects light. This increases the amount of light falling on the visual cells and improves night vision. Because most prosimians

are arboreal, their hands and feet are adapted for grasping (although they are still less dextrous than monkeys and apes). They have flat nails on all digits except the second toe, which instead has a long claw (the “toilet claw”) used for grooming. All species except the aye-aye have a “dental comb”—4 to 6 of the lower front teeth

that are pressed together and grow slightly forward. This is used for mutual grooming.

Movement

Most lemurs are quadrupedal and usually run or leap from branch to branch. Weasel lemurs, the avahi, sifakas, the indri, and galagos are vertical clingers and leapers (see below). On the ground, sifakas and indri move on 2 feet, employing sideways hops with their arms in the air for balance. Lorises and pottos usually clamber slowly along branches, clinging tightly at every step (although when startled they can move swiftly).

Communication

Lemurs produce various calls, both to signal alarm (there are often different calls to distinguish aerial and ground predators) and to communicate within or between troops. Pairs of indri occupy treetop territories, and these are marked with loud wailing calls; small sifaka troops define their areas with “shi-fak” calls that sound like hiccups. Male ring-tailed

TARSIERS

Not strictly prosimians (see the Classification note on p.134), tarsiers are unusual in that they have many primitive features similar to prosimians, yet they also exhibit characteristics linking them to monkeys and apes, such as a dry, hairy nose. The most striking feature of tarsiers is their enormous eyes: each is slightly heavier than the brain. Other characteristics include a large head and ears, long digits with disklike pads at the tips, very long legs with elongated ankles, and a long tail. The 11 species of tarsiers are found in the forests of Southeast Asia, where they spend much of their time clinging to upright tree stems, scanning the forest floor for prey.

lemurs and bamboo lemurs have a wrist gland with a spur, which they use to mark territory by drawing the wrist sharply across a sapling. This produces a click, creates a scar, and leaves a scent—a gesture that is auditory, visual, and olfactory. Galagos leave scent trails around their territory by placing urine on their feet.



VERTICAL LEAP

When leaping from tree to tree, this Verreaux's sifaka uses its long tail for balance, its muscular legs for propulsion, and its large, grasping hands and feet for a secure landing. These features typify prosimians that keep their bodies vertical when climbing. In midair, a semiupright posture is maintained.